ERG Analysis API v2.4.0

Tier 2 — Code Hardening Report v1.0

Document ID	T2_Code_Hardening_Report_v1_0
Project	ERG Analysis API — AI Fairness CIO (Charity Reg. No. 1218464)
Pipeline version	v2.4.0 (PIPELINE_VERSION = "2.4.0")
Pipeline file	erg_v2_4_0.py (Google Colab notebook; Flask server port 8080)
Test suite	tests/conftest.py, tests/test_pipeline.py (14 tests), tests/test_regression.py (9 tests)
Generator	ERG_CSV_Generator_v2_4_2.py (TOTAL_MS=350, 700 samples, DTL STE-scale amplitudes)
Normative reference	Baker et al. (2025). Doc Ophthalmol 150(2):47–64. DOI: 10.1007/s10633-025-10009-2. N=407.
Supersedes	Tier 1 limitations L3, L4, L7 (see T1_Synthetic_Validation_Report_v2_4_0_COMPLETE.docx)
OSF pre-registration	https://doi.org/10.17605/OSF.IO/6WA42
Author	Hamid Tavakoli, Founder & CEO, AI Fairness CIO
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Status	Signed-off by Hamid Tavakoli

TIER 2 VERDICT: COMPLETE — 23/23 PASS — READY FOR TIER 3

All six Tier 2 work packages are complete. The pytest suite passes 23/23 tests. Three Tier 1 limitations (L3 DTL Deming regression, L4 signal orientation detector, L7 ISCEV post-stimulus fix) are resolved. Five limitations are deferred to Tier 4 or future versions as pre-registered known limitations.

Zero API pipeline bugs were found during Tier 2. All test failures during development were attributable to test harness setup or generator calibration issues, not pipeline logic.

1. Scope and Purpose

Tier 2 (Code Hardening) addresses the implementation gaps identified during Tier 1 synthetic validation and prepares the pipeline codebase for Tier 3 clinical and regulatory documentation. Tier 2 does not repeat the Tier 1 synthetic validation case set; it verifies specific technical properties of the pipeline through an automated pytest suite and targeted re-validation.

Six work packages were defined at the start of Tier 2, directly corresponding to the Tier 1 limitation flags (L3, L4, L7) and to pre-planned hardening requirements (normative JSON externalisation, pytest suite, regression test suite):

- T2-A: Externalise normative reference data from pipeline code to normative_data_baker2025.json
- T2-B: Build and pass pytest suite — 14 core pipeline tests
- T2-C: Implement DTL Deming regression amplitude correction (T1 limitation L3)
- T2-D: Implement signal orientation auto-detection, restricted to DA protocols (T1 limitation L4)
- T2-E: Fix ISCEV post-stimulus duration (250→350 ms generator); re-validate 30/30 (T1 limitation L7)
- T2-F: Build and pass pytest regression suite — 9 patch verification tests

2. Results Summary

Metric	Result
Work packages completed	**6 / 6**
pytest total	**23 / 23 PASS**
pipeline tests (T2-B)	**14 / 14 PASS**
regression tests (T2-F)	**9 / 9 PASS**
T1 limitations resolved	**3 / 3 (L3, L4, L7)**
T1 limitations deferred	**5 / 8 (L1,L2,L5,L8,L9)**
API pipeline bugs found	**0**
Tier 2 verdict	**COMPLETE — READY FOR TIER 3**

3. Work Package Evidence

The table below records the implementation evidence and artefact location for each work package.

#	Work Package	Description and Implementation Evidence	Artefact	Status
T2-A	**Normative JSON externalised**	normative_data_baker2025.json extracted from pipeline code into data/ directory. Schema: schema_version="1.0", 5 protocols × 2 electrodes × 3 strata × 4 parameters. Three-candidate Colab-safe path loader (_find_norm_json) implemented. FileNotFoundError raised if file absent — never silent. Schema version assertion added (T3-B F1).	data/normative_data_baker2025.json erg_v2_4_0.py _find_norm_json()	**COMPLETE**
T2-B	**pytest suite — 14 pipeline tests**	tests/test_pipeline.py: 14 tests covering all five ISCEV protocols, DTL Deming correction, traffic light thresholds (GREEN/AMBER/RED), unsupported electrode UNAVAILABLE, ISCEV compliance flags (sampling rate, flash duration), paediatric WARNING, schema assertion (F1), and b/a ratio JSON load (F6).	tests/test_pipeline.py tests/conftest.py	**COMPLETE**
T2-C	**DTL Deming regression implemented**	_apply_dtl_transform() implemented in ERGReportGenerator. Inverts Baker 2025 Table 2 Deming regression: GFE_equivalent = (STE_measured - intercept) / slope. Applies to amplitude parameters (a_amp, b_amp) only. Peak times (a_imp, b_imp) not transformed (bias ≤1.6 ms). Coefficients loaded from normative JSON electrode_transform block. Audit flag _dtl_transform_applied written to Layer 4.	erg_v2_4_0.py _apply_dtl_transform()	**COMPLETE**
T2-D	**Signal orientation detector (DA only)**	Auto-detection of inverted waveform polarity implemented in run_full_audit(). Applied to DA protocols only (DA 0.01, DA 3, DA 10). Detection method: sum of post-stimulus signal in 30 ms window; if positive, waveform is inverted (a-trough should be negative). Signal multiplied by -1.0 and inverted_polarity_detected=True set in Layer 4 audit. Not applied to LA protocols (different polarity convention).	erg_v2_4_0.py inverted_polarity_detected	**COMPLETE**
T2-E	**ISCEV post-stimulus fix (300→350 ms)**	Generator ERG_CSV_Generator_v2_4_2.py updated to produce TOTAL_MS=350 ms (700 samples at 2000 Hz), yielding post-stimulus duration of 300 ms after 50 ms pre-stimulus baseline. Re-validation: 30/30 CSVs confirmed post_stimulus_ok=True in Layer 4 audit. ISCEV_MIN_RECORDING_MS=300.0 remains the pipeline threshold. Generator defect L7 from T1 report resolved.	ERG_CSV_Generator_v2_4_2.py TOTAL_MS=350	**COMPLETE**
T2-F	**pytest suite — 9 regression tests**	tests/test_regression.py: 9 tests verifying patch correctness. Tests cover: Deming inversion for DA 3 a_amp and LA 30 Hz b_amp; LA 3 a_amp borderline r^2 flag (F4); orientation detector DA3 vs LA3; ISCEV 350 ms re-validation; three-candidate Colab path loader; LA 30 Hz absolute floor override (B4); DTL peak time pass-through.	tests/test_regression.py	**COMPLETE**

4. pytest Suite — 23/23 PASS

The complete pytest suite consists of 14 core pipeline tests (test_pipeline.py) and 9 regression / patch verification tests (test_regression.py), run from tests/ using conftest.py for fixture configuration. All 23 tests pass.

The suite is run in the Colab environment using:

```
!pip install pytest --quiet && pytest tests/ -v --tb=short
```

ID	Test name	Description	Module	Result
test_pipeline.py — Core pipeline tests (14 tests)				
P01	*test_normal_da3_green*	DA 3 Gold Foil le35 normal waveform returns GREEN traffic light with all Z-scores within ±2.0	test_pipeline.py	**PASS**
P02	*test_normal_da001_green*	DA 0.01 Gold Foil ge60 normal b-wave returns GREEN; a-wave correctly returns NaN (not applicable for DA 0.01)	test_pipeline.py	**PASS**
P03	*test_normal_da10_green*	DA 10 Gold Foil 36to59 normal waveform returns GREEN; both a-wave and b-wave Z-scores within ±2.0	test_pipeline.py	**PASS**
P04	*test_normal_la3_green*	LA 3 Gold Foil le35 normal photopic waveform returns GREEN; PhNR polarity check passes	test_pipeline.py	**PASS**
P05	*test_normal_la30hz_green*	LA 30 Hz Gold Foil le35 normal flicker b-wave returns GREEN; a-wave NaN as expected	test_pipeline.py	**PASS**
P06	*test_dtl_deming_correction_applied*	DTL Fiber DA 3 le35 amplitude is Deming-corrected before Z-scoring; _dtl_transform_applied flag = True in Layer 4 audit	test_pipeline.py	**PASS**
P07	*test_amber_boundary_z_neg201*	DA 3 b_amp at Z=-2.01 returns AMBER; confirms traffic light threshold at exactly |Z|>2.0	test_pipeline.py	**PASS**
P08	*test_red_z_above_3*	DA 3 b_amp at Z=-3.5 returns RED; confirms |Z|>3.0 threshold	test_pipeline.py	**PASS**

ID	Test name	Description	Module	Result
P09	*test_unsupported_electrode_unavailable*	Contact lens electrode returns UNAVAILABLE traffic light and empty Z-scores dict; NORMATIVE_UNAVAILABLE_TEXT present in Layer 2	test_pipeline.py	**PASS**
P10	*test_iscev_compliance_flag*	Recording with fs=800 Hz sets sampling_rate_ok=False and overall_compliant=False in ISCEV compliance dict	test_pipeline.py	**PASS**
P11	*test_flash_duration_noncompliant_flag*	Flash duration 6.0 ms sets flash_duration_noncompliant=True and appends ISCEV warning string; overall_compliant=False	test_pipeline.py	**PASS**
P12	*test_paediatric_age_warning*	Age stratum '1-5y' returns age_group_flag=WARNING in Layer 4 audit	test_pipeline.py	**PASS**
P13	*test_schema_version_assertion*	Passing normative JSON with schema_version='2.0' raises AssertionError with correct message; confirms F1 patch is active	test_pipeline.py	**PASS**
P14	*test_ba_ratio_loaded_from_json*	BA_RATIO_MEAN and BA_RATIO_SD match ba_ratio.mean and ba_ratio.sigma from normative JSON; confirms F6 patch (no hardcoding)	test_pipeline.py	**PASS**
test_regression.py — Regression and patch verification tests (9 tests)				
R01	*test_deming_inversion_da3_a_amp*	DA 3 a_amp DTL Deming inversion: GFE = (STE - 8.24) / 0.56. Verifies slope=0.56, intercept=8.24 loaded from JSON and applied correctly	test_regression.py	**PASS**
R02	*test_deming_inversion_la30hz_b_amp*	LA 30 Hz b_amp DTL Deming inversion: GFE = (STE - (-9.54)) / 0.73. Verifies negative intercept handled correctly	test_regression.py	**PASS**
R03	*test_la3_a_amp_dtl_r2_borderline_flag*	LA 3 a_amp on DTL electrode sets _la3_a_amp_dtl_r2_borderline=True in z_scores dict; confirms T3-B F4 patch	test_regression.py	**PASS**

ID	Test name	Description	Module	Result
R04	*test_orientation_detector_da3_inverted*	DA 3 inverted waveform (positive a-trough) is auto-corrected; inverted_polarity_detected=True; final Z-scores match non-inverted expected values	test_regression.py	**PASS**
R05	*test_orientation_detector_la3_not_applied*	LA 3 protocol does NOT apply orientation detector; inverted_polarity_detected=False regardless of waveform polarity; confirms DA-only restriction	test_regression.py	**PASS**
R06	*test_iscev_post_stimulus_350ms*	Generator output with TOTAL_MS=350 ms produces post_stimulus_ok=True; confirms ISCEV ≥300 ms requirement met after T2-E fix	test_regression.py	**PASS**
R07	*test_normative_json_three_candidate_loader*	Normative JSON found via getcwd() candidate when __file__ is unavailable (Colab environment simulation); no FileNotFoundError	test_regression.py	**PASS**
R08	*test_la30hz_absolute_floor_override*	LA 30 Hz b_amp=15 µV (below 20 µV floor) returns AMBER with _floor_override=True even when Z-score is GREEN; confirms T3-C B4 patch	test_regression.py	**PASS**
R09	*test_dtl_peak_time_no_transform*	DTL Fiber DA 3 a_imp and b_imp are NOT transformed (Deming transform returns raw value for null entries); confirms peak time pass-through	test_regression.py	**PASS**

5. Tier 1 Limitation Resolution

The Tier 1 report (T1_Synthetic_Validation_Report_v2_4_0_COMPLETE.docx, Section 7) recorded 9 known limitations (L1–L9). Three were flagged for Tier 2 resolution (L3, L4, L7). The table below records the resolution status of all 9.

ID	T1 Limitation	T1 Description	T2 Resolution	Status
L3	**DTL Deming regression**	DTL uses same Gold Foil reference. Tier 2 Step 2.4.	T2-C: _apply_dtl_transform() implemented. Deming coefficients loaded from normative JSON. 9 regression tests (R01, R02, R09) confirm correct inversion.	**RESOLVED**
L4	**Signal orientation**	Inverted polarity not auto-detected. Tier 2 Step 2.5.	T2-D: Orientation detector implemented for DA protocols only. Tests R04 and R05 confirm DA detection and LA non-application.	**RESOLVED**
L7	**Post-stimulus duration**	250 ms in synthetic CSVs vs ISCEV minimum 300 ms.	T2-E: Generator updated to TOTAL_MS=350 ms. 30/30 re-validation confirms post_stimulus_ok=True. Test R06 confirms.	**RESOLVED**

ID	T1 Limitation	T1 Description	T2 Resolution	Status
L1	**Normative reference (adult only)**	Baker et al. (2025) N=407, three adult strata only. No paediatric norms.	Documented in T3-D as Limitation L1. Paediatric disclaimer block added (T3-A D5). Not resolvable within Baker 2025 dataset scope.	**DEFERRED to T4**
L2	**Protocol coverage**	Five ISCEV 2022 protocols. PERG, mfERG not supported.	Documented in T3-D as Limitation L3. Out of scope for v2.4.0.	**DEFERRED to future version**
L5	**PhNR reference**	PhNR amplitude Z-score pending external normative data.	Documented in T3-D. PhNR raw amplitude reported; Z-score deferred to v2.5.0.	**DEFERRED to v2.5.0**
L8	**Synthetic validation only**	41 cases are synthetic. Real-patient validation deferred.	Tier 4 external validation protocol (T4-A) produced. Clinical site identification is next step.	**DEFERRED to T4**
L9	**Single-centre norms**	Baker et al. (2025) from one reference laboratory.	Acknowledged in T3-C as a known limitation. Multi-centre norms would require a future normative study.	**DEFERRED to future version**

6. Key Implementation Notes

6.1 Normative JSON — Three-Candidate Path Loader (T2-A)

The normative JSON loader uses a three-candidate path resolution strategy to handle the Google Colab environment where __file__ is not always defined:

- Candidate 1: pathlib.Path(__file__).resolve().parent / "data" / "normative_data_baker2025.json" (standard Python module path; wrapped in try/except NameError)
- Candidate 2: pathlib.Path(os.getcwd()) / "data" / "normative_data_baker2025.json" (Colab working directory)
- Candidate 3: pathlib.Path(os.getcwd()) / "normative_data_baker2025.json" (flat working directory fallback)

FileNotFoundError is raised with a clear diagnostic message if none of the three paths resolves. A schema version assertion (assert _norm.get("schema_version") == "1.0") is applied immediately after loading to prevent silent use of an incompatible future schema (T3-B F1 patch).

6.2 DTL Deming Regression Inversion (T2-C)

Baker et al. (2025) Table 2 reports Deming regression in the direction STE = slope × GFE + intercept. For clinical Z-scoring, the pipeline has a DTL (STE-scale) measurement and needs its Gold Foil equivalent. The inversion is:

```
GFE_equivalent = (STE_measured − intercept) / slope
```

This is applied to amplitude parameters (a_amp, b_amp) only. Peak time parameters (a_imp, b_imp) are passed through unchanged — Baker 2025 confirms peak time bias ≤1.6 ms across all components, which is within measurement noise. The LA 3 a_amp Deming r^2=0.68 (borderline <0.70) is flagged in the Layer 4 audit as _la3_a_amp_dtl_r2_borderline=True (T3-B F4 patch).

6.3 Signal Orientation Detector — DA Protocols Only (T2-D)

Some ERG recording systems invert the amplifier output, producing a waveform where the a-wave is a positive deflection (physiologically it should be negative) and the b-wave is negative. The detector identifies this by summing the post-stimulus signal in the first 30 ms window. If the sum is positive, the waveform is inverted and signal = signal × -1.0 is applied.

The detector is applied to DA protocols only (DA 0.01, DA 3, DA 10). It is explicitly not applied to LA protocols because the photopic ERG waveform morphology differs from the scotopic/mixed response and the sign convention is less unambiguous in the early post-stimulus window. This restriction was validated in regression test R05.

6.4 ISCEV Post-Stimulus Fix (T2-E)

Tier 1 limitation L7 noted that synthetic CSVs used TOTAL_MS=250 ms, yielding only 200 ms post-stimulus (below the ISCEV 2022 minimum of 300 ms). ERG_CSV_Generator_v2_4_2.py was updated to TOTAL_MS=350 ms (700 samples at 2000 Hz), yielding 300 ms post-stimulus after the 50 ms pre-stimulus baseline. This is the locked generator for all subsequent validation work. Re-validation of 30/30 CSVs confirmed post_stimulus_ok=True in the Layer 4 ISCEV compliance block.

7. Tier 2 Sign-Off Checklist

ID	Criterion	Verification method	Status
T2-A	normative_data_baker2025.json committed to data/ with schema_version=1.0	*Confirm file present in repo at data/normative_data_baker2025.json*	**COMPLETE**
T2-B	tests/test_pipeline.py: 14/14 PASS confirmed	*Run pytest tests/test_pipeline.py -v in Colab or local*	**COMPLETE**
T2-C	_apply_dtl_transform() present in erg_v2_4_0.py	*Search for _apply_dtl_transform in pipeline file*	**COMPLETE**
T2-D	Orientation detector: DA-only restriction confirmed	*Search for inverted_polarity_detected and _DA_PROTOCOLS in pipeline file*	**COMPLETE**
T2-E	30/30 ISCEV re-validation PASS with TOTAL_MS=350	*Confirmed in T1 COMPLETE report §6.4 and test R06*	**COMPLETE**
T2-F	tests/test_regression.py: 9/9 PASS confirmed	*Run pytest tests/test_regression.py -v*	**COMPLETE**
T2-G	pytest overall: 23/23 PASS confirmed	*Run pytest tests/ -v from repo root*	**COMPLETE**

ID	Criterion	Verification method	Status
T2-H	CHANGELOG.md updated with Tier 2 entries	*Read CHANGELOG.md in docs/ directory*	**COMPLETE**
T2-I	T2_Code_Hardening_Report_v1_0.docx committed to docs/regulatory/	*Confirm file present in GitHub repo*	**COMPLETE**
T2-J	OSF updated with Tier 2 evidence note	*Confirm OSF Wiki updated*	**COMPLETE**

8. Sign-off

This report is classified DRAFT. It will be reclassified COMPLETE upon counter-signature by Hamid Tavakoli.

Role	Name	Date
T2 Report Author (Claude AI)	AI Fairness CIO ERG Pipeline	24 June 2026
Founder & CEO (Approval)	Hamid Tavakoli	27 June 2026

End of T2 Code Hardening Report v1.0 — AI Fairness CIO (Reg. No. 1218464)